~~Review of~~ Functional analysis concepts

Disclaimer

These are concepts necessary to understand many aspects of the theory of PDEs as well of the numerical methods of the PDEs.
They are a bit technical, I don't expect you to remember everything, but to understand why we need to introduce some concepts, in particular, when we will use them in the future lectures.

Functional spaces

We are used to think about vector spaces as $\mathbb{R}^n$ or $\mathbb{C}^n$. However, we can also think about spaces of functions as vector spaces. For example, the space of continuous functions on an interval $[a, b]$ is a vector space, since the sum of two continuous functions is continuous and the scalar multiplication of a continuous function is also continuous.
We will call them **functional spaces** as their elements are functions.

Examples of (normed) functional spaces

- $C(\Omega)$: the space of continuous functions on Ω, with the supremum norm $||f||_\infty = \sup_{x\in\Omega} |f(x)|$.
- $C^k(\Omega)$: the space of functions on Ω that are k-times continuously differentiable, with the norm $||f||_{C^k} = \sum_{p_1+\cdots+p_d\leq k} ||\partial_{x_1^{p_1}\ldots x_d^{p_d}} f||_\infty$.
- $C^\infty(\Omega)$: the space of functions on Ω that are infinitely differentiable.
- $L^2(\Omega) := \{f : \Omega \to \mathbb{R} : \int_\Omega |f(x)|^2 dx < \infty\}$: the space of square-integrable functions on Ω, with the norm $||f||_{L^2} = \sqrt{\int_\Omega |f(x)|^2 dx}$.

Keep these examples in mind, we will see them again and again!

Linear and bilinear functionals

Given a functional space V, a **linear functional** is a map $L : V \to \mathbb{R}$ that satisfies linearity: $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$ for all $u, v \in V$ and scalars $\alpha, \beta \in \mathbb{R}$.

A **bilinear functional** is a map $B : V \times V \to \mathbb{R}$ that is linear in each argument.

Examples

- The evaluation functional $L_{x_0} : C(\Omega) \to \mathbb{R}$ defined by $L_{x_0}(u) := u(x_0)$ for a fixed $x_0 \in \Omega$ is a linear functional;
- The integral functional $L : L^2(\Omega) \to \mathbb{R}$ defined by $L(u) := \int_\Omega u(x)dx$ is a linear functional;
- The bilinear form $B : L^2(\Omega) \times L^2(\Omega) \to \mathbb{R}$ defined by $B(u, v) := \int_\Omega u(x)v(x)dx$ is a bilinear functional.

Linear and bilinear functionals properties

Boundedness and Continuity

A functional L is bounded if there exists a constant C such that $|L(u)| \leq C||u||_V$ for all $u \in V$. If V is a Banach space (normed and complete), then a linear bounded functional is also continuous.

Example of unbounded functional

The "evaluation in 0 of the derivative" functional $L : C^\infty([0,1]) \to \mathbb{R}$ defined by $L(u) := u'(0)$ is not bounded with respect to the supremum norm, since for any $C > 0$, we can find a function u such that $||u||_\infty = 1$ but $|L(u)| = |u'(0)| \to \infty$, e.g.

$$u_n(x) = \sin(n\,x) \implies ||u_n||_\infty = 1, \quad L(u_n) = u_n'(0) = n \to \infty.$$

Dual Space

The dual space $V^* = V'$ is the space of all bounded linear functionals on V.

$$V^* := \{F : V \to \mathbb{R} : \ F \text{ is linear and bounded}\}.$$

Norm

The norm of a functional $L \in V^*$ is defined as
$||L||_{V^*} = \sup_{||u||_V \leq 1} |L(u)| = \sup_{||u||_V \neq 0} \frac{|L(u)|}{||u||_V}$.

Hilbert Space

A Hilbert space H is a real or complex inner product space that is also a complete metric space with respect to the distance function induced by the inner product.

The inner product is a bilinear function $(\cdot,\cdot)_H : V \times V \to \mathbb{R}$ that is symmetric and positive definite. The induced norm is $||u||_H := \sqrt{(u,u)_H}$.

Example

$L^2(\Omega)$ is a Hilbert space with the inner product defined as $(f,g)_{L^2} = \int_\Omega f(x)g(x)dx$.

Riesz Representative

The **Riesz representation theorem** states that for every bounded linear functional L on a Hilbert space H, there exists a unique element $v_L \in H$ such that

$$L(u) = (u, v_L)_H$$

for all $u \in H$. Moreover, $||L||_{H^*} = ||u_L||_H$.
Conversly, for every element $u \in H$ there exists a linear and bounded functional L_u such that

$$L_u(v) = (u, v)_H \text{ for every } v \in H.$$

Moreover, $||L_u||_{H^*} = ||u||_H$.

Hence, there is a bijection between H and H^*.

Bilinear form definitions

Given V a normed functional space, a bilinear form a is a function that maps every two elements of V to a scalar

$$a : V \times V \to \mathbb{R}.$$

A form

- is **bilinear** if
  - $a(\lambda u + \mu w, v) = \lambda a(u, v) + \mu a(w, v)$ for every $\lambda, \mu \in \mathbb{R}$ and every $v, w, u \in V$, and
  - $a(u, \lambda v + \mu w) = \lambda a(u, v) + \mu a(u, w)$ for every $\lambda, \mu \in \mathbb{R}$ and every $v, w, u \in V$;
- is **continuous** if there exists an $M > 0$ such that

$$a(u, v) \leq M \|u\|_V \|v\|_V \text{ for every } v, u \in V;$$

- is **symmetric** if $a(u, v) = a(v, u)$ for every $u, v \in V$;
- is **positive** if $a(v, v) > 0$ for all $v \in V$ with $v \neq 0$;
- is **coercive** if there exists $\alpha > 0$ such that $a(v, v) > \alpha \|v\|_V^2$ for all $v \in V$.

Distributions

Let $\Omega \subset \mathbb{R}^d$ be an open set and $f : \Omega \to \mathbb{R}$ a function.

Support of a Function

The support of a function f, denoted by $\mathrm{supp}(f)$, is the closure of the set where f is non-zero.

$$\mathrm{supp}(f) := \overline{\{x \in \Omega : f(x) \neq 0\}}.$$

Compact Support

A function has compact support if its support is a compact set (compact = closed and bounded for us).

C^∞ Compact Support Functions

A function is in $\mathcal{D}(\Omega) := C_c^\infty(\Omega)$ if it is infinitely differentiable and has compact support in Ω.

Convergence in $\mathcal{D}(\Omega)$

A sequence of functions $\{f_n\}$ in $\mathcal{D}(\Omega)$ converges to f in $C_c^\infty(\Omega)$ if

- exists a fixed compact set K that contains all supports of f_n
- all derivatives of f_n converge uniformly to the corresponding derivatives of f, i.e. $\partial_{x_1^{p_1}\dots x_d^{p_d}} f_n \to \partial_{x_1^{p_1}\dots x_d^{p_d}} f$ for all $p_1, \dots, p_d$.

Distributions

A **distribution** is a linear functional $T : \mathcal{D}(\Omega) \to \mathbb{R}$ that is continuous, i.e.,

$$\lim_{k\to\infty} T(\varphi_k) = T(\varphi),$$

for all $\varphi_k \to_{\mathcal{D}} \varphi \in \mathcal{D}$.
Hence, the distribution space $\mathcal{D}^*(\Omega)$ is the dual of $\mathcal{D}(\Omega)$.

Notation for distribution $T \in \mathcal{D}^*(\Omega)$ applied to a function $f \in \mathcal{D}(\Omega)$: $T(f) = \langle T, f\rangle$.

Example Dirac Delta

The Dirac delta distribution δ_a with $a \in \Omega$ a point, is defined by $\delta_a(\phi) = \phi(a)$ for all $\phi \in \mathcal{D}(\Omega)$. It is a distribution that "picks out" the value of a function at a point.
The Dirac delta is not a functional on the $L^2(\mathbb{R})$ space, since it is not bounded with respect to the L^2 norm.
However, it is a distribution in $\mathcal{D}^*(\Omega)$.

Convergence in $\mathcal{D}^*(\Omega)$

A sequence of distributions T_n converges in $\mathcal{D}^*(\Omega)$ to $T \in \mathcal{D}^*(\Omega)$ if

$$\lim_{n\to\infty} T_n(\varphi) = T(\varphi), \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

$L^2(\Omega)$ squared summable functions

$$L^2(\Omega) := \{f : \Omega \to \mathbb{R} \text{ such that } \int_\Omega f(x)^2 \mathrm{d}x < \infty\}.$$

1. $L^2(\Omega)$ is a Hilbert space with scalar product $(f, g) := \int_\Omega f(x)g(x)\mathrm{d}x$.
2. The $L^2(\Omega)$ norm is define through the inner product as $\|f\|_{L^2(\Omega)} := \sqrt{\int_\Omega f(x)^2 \mathrm{d}x}$.
3. To every function $f \in L^2(\Omega)$ is associated a distribution $T_f \in \mathcal{D}^*(\Omega)$ defined by

$$T_f(\varphi) := \int_\Omega f(x)\varphi(x)\mathrm{d}x, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

4. $\mathcal{D}(\Omega)$ is dense in $L^2(\Omega)$, i.e., for every $f \in L^2(\Omega)$ there exists a sequence of $\varphi_k \in \mathcal{D}(\Omega)$ such that

$$\|\varphi_k - f\|_{L^2(\Omega)} \to 0.$$

5. $\mathcal{D}(\Omega) \subset L^2(\Omega) \Longrightarrow (L^2(\Omega))^* = L^2(\Omega) \subset \mathcal{D}^*(\Omega)$.

Example: convergence to Dirac Delta

Let $\chi_{[a,b]}$ be the characteristic function on the interval $[a,b] \subset \mathbb{R}$ defined as

$$\chi_{[a,b]}(x) = \begin{cases} 0 & \text{if } x \notin [a,b], \\ 1 & \text{if } x \in [a,b]. \end{cases}$$

Let us build the sequence of functions in $L^2(\mathbb{R})$ $f_n(x) := \frac{n}{2}\chi_{[-1/n,1/n]}(x)$. Clearly, we have that

1. $\int_{\mathbb{R}} f_n(x)\mathrm{d}x = 1$
2. $T_{f_n}(\varphi) = \int_{\mathbb{R}} f_n(x)\varphi(x)\mathrm{d}x = \frac{n}{2}\int_{-1/n}^{1/n} \varphi(x)\mathrm{d}x = \frac{n}{2}(\Phi(1/n) - \Phi(-1/n))$ where $\frac{d}{dx}\Phi(x) = \varphi(x)$.
3. Let $h_n = 1/n$, $T_{f_n}(\varphi) = \frac{\Phi(h)-\Phi(-h)}{2h}$
4. $\lim_{n\to\infty} T_{f_n}(\varphi) = \lim_{n\to\infty} \frac{\Phi(h)-\Phi(-h)}{2h} = \frac{d}{dx}\Phi(0) = \varphi(0)$.
5. $T_{f_n}(\varphi) \to \varphi(0) = \delta_0(\varphi)$.
6. δ_0 is a distribution in $\mathcal{D}^*(\mathbb{R})$ but not a function in $L^2(\mathbb{R})$.

Definition: Derivation in distributional sense

Let $T \in \mathcal{D}^*(\Omega)$, with $\Omega \subset \mathbb{R}^d$ open. We can define the derivative of T using the integration by parts.

$$\partial_{x_i} T(\varphi) = \langle \partial_{x_i} T, \varphi \rangle := -\langle T, \partial_{x_i} \varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

If T is a T_f with $f \in \mathcal{C}^1(\Omega)$, it is clearly the classical derivative. Let's see in 1D with $\Omega = [a, b]$.

$$\partial_x T_f(\varphi) = \langle \partial_x T_f, \varphi \rangle = \int_a^b \partial_x f(x) \varphi(x) \mathrm{d}x = \underbrace{[f(x)\varphi(x)]_a^b}_{=0} - \int_a^b f(x) \partial_x \varphi(x) \mathrm{d}x, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

But the definition holds for every distribution, even for discontinuous functions, for the Dirac delta, etc. !!!

Higher derivatives

$$\left\langle \frac{\partial^{p_1 + \cdots + p_d} T}{\partial x_1^{p_1} \ldots \partial x_d^{p_d}}, \varphi \right\rangle := (-1)^{p_1 + \cdots + p_d} \left\langle T, \frac{\partial^{p_1 + \cdots + p_d} \varphi}{\partial x_1^{p_1} \ldots \partial x_d^{p_d}} \right\rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega) = C_c^\infty(\Omega).$$

Example: Derivative of Heaviside function

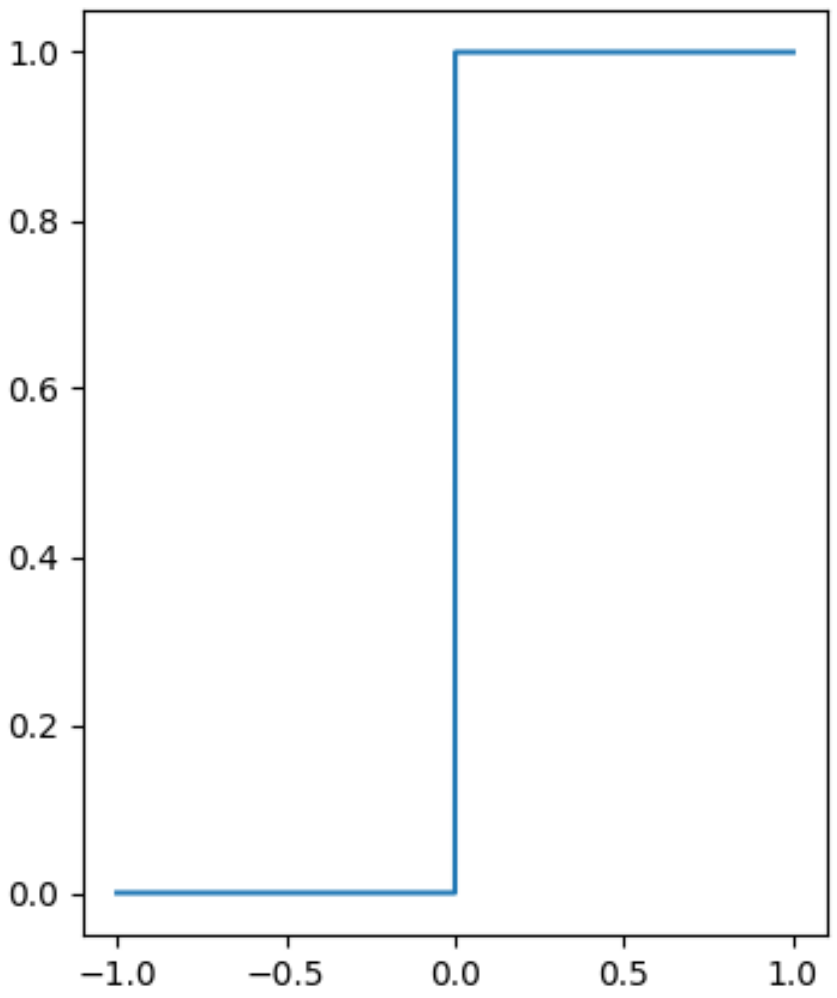
$$R(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x \leq 0, \end{cases}$$

- $R \in L^2((-1,1))$
- $R \notin C((-1,1))$
- $T_R \in \mathcal{D}^*((-1,1))$

$$\langle \partial_x T_R, \varphi \rangle = -\int_{-1}^{1} R(x)\partial_x \varphi(x)\mathrm{d}x$$

$$= -\int_{0}^{1} \partial_x \varphi(x)\mathrm{d}x = -[\varphi]_0^1 = \varphi(0)$$

$$\implies \partial_x R = \delta_0.$$



Sobolev Spaces

As we have seen $L^2(\Omega) \subset \mathcal{D}^*(\Omega)$. This does not imply that their distributional derivatives are still in L^2. The Heaviside function is in L^2 but its derivative it's not.

We need to introduce other spaces to distinguish functions!

Sobolev spaces

Let $\Omega \subset \mathbb{R}^d$ and $k \in \mathbb{N}_0$. We define the Sobolev space of order k on Ω the space of the functions in $L^2(\Omega)$ with distributional derivatives up to order k in $L^2(\Omega)$.

$$H^k(\Omega) := \{f \in L^2(\Omega) : \partial_{x_1^{p_1} \dots x_d^{p_d}} f \in L^2(\Omega), \text{ for all } p_1, \dots, p_d : p_1 + \dots + p_d \leq k\}.$$

- $H^{k+1}(\Omega) \subset H^k(\Omega)$
- $L^2(\Omega) = H^0(\Omega)$
- Heaviside $R \in H^0((-1,1))$, but $R \notin H^1((-1,1))$

Examples

- Example of $H^\infty(\Omega)$ but not $C(\Omega)$

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 0, \\ 3 & \text{if } x = 0. \end{cases}$$

- Example of $H^1(\Omega)$ but not $H^2(\Omega)$

$$f(x) = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases} \qquad f'(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases}$$

Norms and inner products of Sobolev spaces

- Sobolev spaces $H^k(\Omega)$ are Hilbert space with respect to the following scalar product

$$(f,g)_k = (f,g)_{H^k(\Omega)} := \sum_{p_1+\cdots+p_d\le k} \int_\Omega \partial_{x_1^{p_1}\ldots x_d^{p_d}} f \cdot \partial_{x_1^{p_1}\ldots x_d^{p_d}} g \,\mathrm{d}x,$$

  with the norms

$$\|f\|_k = \|f\|_{H^k(\Omega)} := \sqrt{\sum_{p_1+\cdots+p_d\le k} \int_\Omega (\partial_{x_1^{p_1}\ldots x_d^{p_d}} f)^2 \,\mathrm{d}x},$$

- Seminorms

$$|f|_k = |f|_{H^k(\Omega)} := \sqrt{\sum_{p_1+\cdots+p_d= k} \int_\Omega (\partial_{x_1^{p_1}\ldots x_d^{p_d}} f)^2 \,\mathrm{d}x},$$

- $\|f\|_k = \sqrt{\sum_{m=0}^k |f|_m^2}$

Examples for $k = 1$

$$(f, g)_1 = (f, g)_{H^1(\Omega)} = \int_\Omega f(x)\, g(x)\, \mathrm{d}x + \int_\Omega f'(x)\, g'(x)\, \mathrm{d}x$$

$$\|f\|_1 = \sqrt{\int_\Omega f^2(x)\, \mathrm{d}x + \int_\Omega (f'(x))^2\, \mathrm{d}x} = \sqrt{\|f\|^2_{L^2(\Omega)} + \|f'\|^2_{L^2(\Omega)}}$$

$$|f|_1 = \sqrt{\int_\Omega (f'(x))^2\, \mathrm{d}x} = \|f'\|_{L^2(\Omega)}$$

Boundary for bounded domains

Property

If $\Omega \subset \mathbb{R}^d$ is open with a *smooth enough* boundary, then $H^k(\Omega) \subset C^m(\bar{\Omega})$ if $m < k - \frac{d}{2}$.

Careful, in this case we mean that there is a representative of the function in H^k such that it also belongs to C^m. In the previous example where

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 0, \\ 3 & \text{if } x = 0. \end{cases}$$

there exists a continuous representative of this function $f(x) = x^2$ which is the same function in $L^2(\Omega)$.

$H_0^1(\Omega)$

Let Ω be a bounded domain. We denote with $H_0^1(\Omega)$ the closure of $\mathcal{D}(\Omega)$ in $H^1(\Omega)$. (*morally zero on the boundary*)

Poincarè inequality

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with a Lipschitz boundary. There exists a constant $C = C(\Omega) > 0$ such that for all $u \in H_0^1(\Omega)$,

$$\|u\|_{L^2(\Omega)} \leq C\|\nabla u\|_{L^2(\Omega)} = C|u|_1.$$

Proof

Since $\Omega \subset \mathbb{R}^d$ is bounded there exists a ball $S_R = \{x : |x - x_0| < R\}$ that contains Ω. Since, $\mathcal{D}(\Omega)$ is dense in $H_0^1(\Omega)$, we can prove the inequality for $u \in \mathcal{D}(\Omega)$ and pass to the limit to get it for H_0^1. Notice that $\operatorname{div}(x - x_0) = d$. So, using Cauchy-Schwarz,

$$\|u\|_{L^2(\Omega)}^2 = d^{-1}\int_\Omega d\,|u(x)|^2 \mathrm{d}x = d^{-1}\int_\Omega \operatorname{div}(x - x_0)|u(x)|^2 \mathrm{d}x = -d^{-1}\int_\Omega (x - x_0)\nabla(|u(x)|^2)\mathrm{d}x =$$

$$-2d^{-1}\int_\Omega (x - x_0)u(x)\nabla(u(x))\mathrm{d}x \leq 2d^{-1}\|x - x_0\|_\infty \|u\|_{L^2(\Omega)}\|\nabla u\|_{L^2(\Omega)} = 2d^{-1}R\|u\|_{L^2(\Omega)}|u|_1.$$

Proposition

On $H_0^1(\Omega)$ the seminorm $|\cdot|_1$ is actually a norm and it is equivalent to $\|\cdot\|_1$.

Proof

$$\|u\|_1^2 = |u|_1^2 + \|u\|_{L^2}^2 \leq (1 + C^2)|u|_1^2.$$

On the other hand

$$|u|_1^2 \leq |u|_1^2 + \|u\|_{L^2}^2 = \|u\|_1^2.$$